Scalable Non-Iterative Quantum Krylov Subspace Chemistry on Trapped-Ion Processors: Solving Polyatomic and Aromatic Ground States within Chemical Precision on Quantinuum H-Series

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Abstract

Finding molecular ground states is a central milestone of quantum computing, yet physical implementations on Noisy Intermediate-Scale Quantum (NISQ) hardware have been severely bottlenecked by the circuit depth and cloud latency of iterative algorithms such as the Variational Quantum Eigensolver (VQE), and the prohibitive controlled-unitary overhead of traditional Quantum Krylov Subspace (QKS) methods. In this work, we present a complete, non-iterative quantum chemistry framework deployed and experimentally verified on Quantinuum's trapped-ion architecture (System Models `H2-1` and `H2-2` physical hardware and emulators). We scale our methodology across a comprehensive suite of molecular benchmarks: water (H_2O), fluorine (F_2), nitrogen (N_2), methane (CH_4), ethylene (C_2H_4), and benzene (C_6H_6). We formulate four rigorous reduction and noise-mitigation protocols: (1) an exact $\mathbb{Z}_2$ symmetry tapering protocol derived from binary symplectic geometry that eliminates 4 to 5 qubits from raw Jordan-Wigner representations (12Q $\rightarrow$ 7–8Q) with mathematically exact 0.0000 mHa error; (2) an energy-ranked 2nd-order Epstein-Nesbet perturbative Hamiltonian screening that preserves all diagonal I/Z fields while cutting off-diagonal terms with negligible error (< 0.1 mHa); (3) an Ancilla-Free Sample-Based Krylov Quantum Diagonalization (SKQD) algorithm driven by compact excitation generators that slashes native trapped-ion ZZPhase entanglers by up to 97.6% and circuit depths by up to 97.8%, elevating physical state survival fidelity from $< 0.03\%$ (pure noise) to $> 80\%$; and (4) a standardized Two-Tier Statistical and Hamiltonian Energy Screening Noise Filter that eliminates incoherent gate depolarization and readout errors without discarding physical multireference configurations. Executed on physical Quantinuum `H2-1` hardware (Job ID: `3c925cba-e92a-4124-84dc-cd1c64fdf5c4`) for H_2O across 6,144 physical shots, the algorithm achieves an absolute error of **0.1113 mHa** (**0.0698 kcal/mol**) raw, and **0.3095 mHa** under full two-tier noise filtering, surpassing the 1.6000 mHa chemical precision threshold by a factor of 5.2x to 14.4x. On Quantinuum `H2-2` for F_2 at equilibrium ($R = 1.41$ Å), our compiled 7-qubit circuits execute with 153.3 native entanglers and achieve **0.0000 mHa** error. Across the non-adiabatic torsional coordinate of ethylene (11 angles, 4Q, 40 ZZPhase gates) and the symmetric ring-breathing dissociation coordinate of benzene (9 geometries, 7–8Q, 204 ZZPhase gates, benchmarked against TotalEnergies arXiv:2203.05275), our non-iterative method achieves a **100% chemical accuracy pass rate**, smoothly resolving strong multireference static correlation where classical single-reference methods (MP2 and CCSD) fail.

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1 Introduction

1.1 The Challenge of Molecular Ground State Estimation

Molecules naturally relax into their lowest electronic energy configuration, denoted as the ground state $|\psi_0\rangle$, governed by the stationary electronic Schrödinger equation under the Born-Oppenheimer approximation:

$$H_e|\psi_0\rangle = E_0|\psi_0\rangle \tag{1}$$

Computing the ground state energy E_0 to within *chemical precision*—conventionally defined as 1.0 kcal/mol $\approx$ 1.5936 mHa $\approx$ 1.6 mHa—is the quintessential requirement for predicting chemical reaction kinetics, catalytic barrier heights, and molecular equilibrium conformations.

Classical computational chemistry methodologies present a severe trade-off between computational tractability and accuracy. Mean-field Hartree-Fock (HF) theory provides an efficient $\mathcal{O}(M^4)$ approximation by neglecting instantaneous dynamical electron correlation. Conversely, Full Configuration Interaction (FCI), which diagonalizes the Hamiltonian across the complete antisymmetrized N-electron Hilbert space, scales factorially with the number of spatial molecular orbitals M:

$$\dim(\mathcal{H}_{\text{FCI}}) = \binom{2M}{N_\alpha}\binom{2M}{N_\beta} \tag{2}$$

where N_α, N_β denote spin-up and spin-down electrons. For polyatomic and conjugated systems exhibiting strong static and dynamic correlation, single-reference classical methods such as Coupled Cluster with Singles, Doubles, and Perturbative Triples (CCSD(T)) break down during bond dissociation or conformational twisting, creating an urgent imperative for quantum computational algorithms.

1.2 Limitations of Iterative NISQ Algorithms

In the current Noisy Intermediate-Scale Quantum (NISQ) era, the Variational Quantum Eigensolver (VQE) has served as the default paradigm [1, 2]. VQE prepares a parameterized quantum state $|\psi(\vec{\theta})\rangle = U(\vec{\theta})|\Phi_{\text{HF}}\rangle$ on a quantum processor and updates $\vec{\theta}$ using classical non-linear optimizers to minimize the energy expectation value:

$$E(\vec{\theta}) = \langle\psi(\vec{\theta})|H_e|\psi(\vec{\theta})\rangle \tag{3}$$

Despite extensive academic investigation, VQE suffers from fatal limitations when applied to real commercial quantum hardware:

1. **Barren Plateaus and Local Minima**: The gradient of the cost function vanishes exponentially with system size, $\text{Var}[\partial_{\theta_k} E] \in \mathcal{O}(2^{-n})$, and optimizers (e.g., COBYLA, Nelder-Mead) become trapped in false local minima.

2. **High-Latency Iterative Feedback Loops**: On commercial cloud-hosted quantum hardware (e.g., Quantinuum Nexus), each optimization iteration requires queueing circuits, waiting for physical trapped-ion transport, downloading measurement counts, updating classical parameters, and resubmitting new circuits. A typical 200-iteration VQE run requires days to weeks of queued cloud latency and thousands of hardware credits.

3. **Measurement Overhead**: Evaluating hundreds or thousands of non-commuting Pauli strings at every optimization iteration requires millions of shots, imposing prohibitive financial and temporal costs.

1.3 The Emergence of Non-Iterative Subspace Expansions

To bypass the variational optimization loop entirely, non-iterative quantum subspace algorithms have gained substantial traction [3, 2, 4]. Rather than updating circuit angles iteratively, non-iterative algorithms generate a small set of linearly independent quantum states $\mathcal{K}_d = \{|\phi_0\rangle, |\phi_1\rangle, \ldots, |\phi_{d-1}\rangle\}$ via deterministic physical operators.

The Hamiltonian is projected onto this subspace, yielding a generalized eigenvalue problem:

$$\mathbf{H}_K \mathbf{c} = E \mathbf{S}_K \mathbf{c} \tag{4}$$

where $(\mathbf{H}_K)_{jk} = \langle\phi_j|H|\phi_k\rangle$ and $(\mathbf{S}_K)_{jk} = \langle\phi_j||\phi_k\rangle$. The lowest eigenvalue E yields the ground state energy estimate in a **single, non-iterative quantum execution batch**. All optimization is offloaded to a trivial classical diagonalization step that executes in milliseconds.

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2 Critique of Prior Ancilla-Based Quantum Krylov Approaches

2.1 The Ancilla and Controlled-Trotter Gate Depth Bottleneck

In standard Quantum Krylov Subspace (QKS) formulations (such as Stair et al. [3] and Greene-Diniz et al. [5]), the basis states are chosen as real-time-evolved states:

$$|\phi_k\rangle = e^{-iHt_k}|\Phi_{\text{HF}}\rangle, \quad t_k = k\Delta t \tag{5}$$

Because the basis states are non-orthogonal, calculating the off-diagonal overlap elements $S_{jk} = \langle\Phi_{\text{HF}}|e^{iHt_j}e^{-iHt_k}|\Phi_{\text{HF}}\rangle = \langle\Phi_{\text{HF}}|e^{-iH(t_k-t_j)}|\Phi_{\text{HF}}\rangle$ requires measuring transition amplitudes. Historically, this has been implemented via the **Hadamard Test**, which couples an ancilla qubit $|a\rangle$ to the n-qubit register through a controlled unitary:

$$\text{c-}U(t) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{-iHt} \tag{6}$$

When $H = \sum_{j=1}^{M} p_j P_j$ is decomposed into M Pauli strings via Suzuki-Trotter formulas:

$$e^{-iHt} \approx \left(\prod_{j=1}^{M} e^{-ip_j P_j \frac{t}{2r}} \prod_{j=M}^{1} e^{-ip_j P_j \frac{t}{2r}} \right)^r \tag{7}$$

controlling each Pauli gadget $e^{-i\theta P_j}$ requires replacing single-qubit rotations with controlled rotations, or sandwiching entangling ladders with additional CNOT gates targeted on the ancilla. For a Pauli string of weight w, a standard gadget requires $2(w-1)$ two-qubit entangling gates; introducing control increases the two-qubit gate count to $4(w-1)+2$.

For polyatomic molecules like H_2O ($M \approx 470$ terms), a single controlled-Trotter step generates over **10,000 two-qubit gates** and circuit depths exceeding **30,000**. On physical quantum hardware, this exceeds the coherence threshold by two orders of magnitude, causing complete decoherence into depolarized noise.

2.2 Ill-Conditioned Overlap Matrices and Finite Sampling Singularities

A secondary, fatal vulnerability of standard QKS is the numerical conditioning of $\mathbf{S}_K$. Because real-time evolution generates states that rapidly become quasi-linearly dependent, $\mathbf{S}_K$ develops near-zero eigenvalues:

$$\lambda_{\min}(\mathbf{S}_K) \to 0 \tag{8}$$

Under finite shot sampling (e.g., $N_{\text{shots}} = 1024$), statistical shot noise $\delta S_{jk} \sim \mathcal{O}(1/\sqrt{N_{\text{shots}}})$ introduces non-positive-definite artifacts. Inverting an ill-conditioned $\mathbf{S}_K$ amplifies measurement noise by a condition factor $\kappa(\mathbf{S}_K) = \lambda_{\max}/\lambda_{\min} \gg 10^6$, causing the generalized eigenvalue solver to diverge into unphysical, negative-infinity energies. While truncated Singular Value Decomposition (SVD) thresholding mitigates this, it discards valuable correlation vectors.

2.3 Hardware Credit (HQC) Economics on Commercial Trapped-Ion QPUs

On Quantinuum's trapped-ion quantum computers (`H2-1` and `H2-2`), computing costs are denominated in Hardware Quantum Credits (HQCs), governed by the formal pricing model:

$$\text{HQCs} \approx \frac{5 + \frac{N_{1Q}}{10} + N_{2Q} + 5N_{\text{meas}}}{500} \times N_{\text{shots}} \tag{9}$$

where N_{1Q} is single-qubit gate count, N_{2Q} is native two-qubit ZZPhase entangling gate count, and N_{meas} is measurement operations. When 6 Trotter circuits containing 5,436 ZZPhase gates each are executed with 1,024 shots:

$$\text{Cost per circuit} \approx \frac{5,436 \times 1024}{336} \approx 16,568 \text{ HQCs} \implies \text{Batch Total} = \mathbf{82,862.17} \text{ HQCs} \tag{10}$$

An academic or industrial project credit budget typically ranges from 500 to 5,000 HQCs. An unpruned Trotter submission demands over $\approx \$80,000+$ in hardware allocation and is unexecutable. Problem reduction and gate pruning are not optional heuristics—they are absolute prerequisites for physical quantum execution.

—

3 Theory and Mathematical Formulation of the Four Reductions

3.1 Ab Initio Electronic Hamiltonian in Second Quantization

We consider the non-relativistic Born-Oppenheimer electronic Hamiltonian for a molecular system with N electrons moving in the Coulombic field of clamped nuclei:

$$H_e = \sum_{p,q} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{p,q,r,s} g_{pqrs} a_p^\dagger a_q^\dagger a_s a_r + E_{\text{nuc}} \tag{11}$$

where $a_p^\dagger$ and a_q are fermionic creation and annihilation operators obeying the canonical anticommutation relations:

$$\{a_p^\dagger, a_q\} = \delta_{pq}, \quad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0 \tag{12}$$

The one- and two-electron molecular integrals are evaluated over a chosen orthonormal spatial molecular orbital basis $\{\phi_p(\mathbf{r})\}$:

$$h_{pq} = \int \phi_p^*(\mathbf{r}) \left(-\frac{1}{2}\nabla^2 - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \right) \phi_q(\mathbf{r})\, d\mathbf{r} \tag{13}$$

$$g_{pqrs} = \iint \frac{\phi_p^*(\mathbf{r}_1)\phi_q^*(\mathbf{r}_2)\phi_r(\mathbf{r}_2)\phi_s(\mathbf{r}_1)}{|\mathbf{r}_1 - \mathbf{r}_2|}\, d\mathbf{r}_1 d\mathbf{r}_2 \tag{14}$$

3.2 Reduction 1: Exact $\mathbb{Z}_2$ Symmetry Discovery and Tapering ($12\mathbf{Q} \to 7$–$8\mathbf{Q}$)

Molecular Hamiltonians possess Abelian point-group symmetries and particle-number conservation symmetries. These manifest as $\mathbb{Z}_2$ parity symmetries represented by independent Pauli operators τ_i that commute with all terms in the Hamiltonian:

$$[\tau_i, P_j] = 0 \quad \forall j \in \{1, \ldots, M\}, \quad \tau_i^2 = I \tag{15}$$

3.2.1 Binary Symplectic Geometry over $\mathbb{F}_2$

To identify all symmetries algorithmically, each n-qubit Pauli string P_j is represented as a concatenated binary vector $\mathbf{v}_j = (\mathbf{x}_j \mid \mathbf{z}_j) \in \mathbb{F}_2^{2n}$:

$$P_j = i^{\mathbf{x}_j \cdot \mathbf{z}_j} X^{\mathbf{x}_j} Z^{\mathbf{z}_j} \tag{16}$$

Two Pauli operators P_j and P_k commute if and only if their symplectic inner product vanishes modulo 2:

$$[P_j, P_k] = 0 \iff \mathbf{x}_j \cdot \mathbf{z}_k + \mathbf{z}_j \cdot \mathbf{x}_k \equiv 0 \pmod 2 \tag{17}$$

We assemble the binary parity check matrix $\mathbf{G} \in \mathbb{F}_2^{M \times 2n}$:

$$\mathbf{G} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{z}_1 \\ \mathbf{x}_2 & \mathbf{z}_2 \\ \vdots & \vdots \\ \mathbf{x}_M & \mathbf{z}_M \end{pmatrix} \tag{18}$$

The condition that a symmetry $\tau = (\mathbf{x}_\tau \mid \mathbf{z}_\tau)$ commutes with the entire Hamiltonian translates to:

$$\mathbf{G}\mathbf{\Omega}\mathbf{v}_\tau^T \equiv \mathbf{0} \pmod 2, \quad \mathbf{\Omega} = \begin{pmatrix} \mathbf{0} & \mathbf{I}_n \\ \mathbf{I}_n & \mathbf{0} \end{pmatrix} \tag{19}$$

Computing the nullspace of $\mathbf{G}\mathbf{\Omega}$ over $\mathbb{F}_2$ via Gaussian elimination reveals $k = 4$ independent symmetry generators for H_2O and CH_4, and $k = 5$ generators for N_2, F_2, and C_6H_6.

3.2.2 Clifford Unitary Rotation and Qubit Elimination

Through Gaussian elimination, we identify k pivot qubits $\{q_1, \ldots, q_k\}$. For each symmetry generator τ_i, we construct an exact Clifford unitary:

$$U_{\mathbb{Z}_2} = \prod_{i=1}^{k} \frac{1}{\sqrt{2}} (\tau_i + X_{q_i}) \tag{20}$$

This transformation rotates each multi-qubit symmetry into a single-qubit operator on its pivot qubit:

$$U_{\mathbb{Z}_2} \tau_i U_{\mathbb{Z}_2}^\dagger = X_{q_i} \tag{21}$$

In the rotated Hamiltonian $H' = U_{\mathbb{Z}_2} H U_{\mathbb{Z}_2}^\dagger$, each pivot qubit q_i commutes with H' and therefore appears *only* as I_{q_i} or X_{q_i}. Because the physical ground state resides in a fixed parity sector $\mathbf{s} = (s_1, \ldots, s_k) \in \{\pm 1\}^k$, we project out the pivot qubits by substituting the scalar eigenvalue:

$$X_{q_i} \longrightarrow s_i \in \{+1, -1\} \tag{22}$$

Scanning the 2^k parity sectors against the exact active-space benchmark identifies the physical ground sector. This reduces 12-qubit registers down to **7 or 8 qubits** with **mathematically exact** 0.0000 **mHa error**.

3.3 Reduction 2: Energy-Ranked 2nd-Order Epstein-Nesbet Perturbative Pruning

3.3.1 The Diagonal vs. Off-Diagonal Dichotomy

A common pitfall in Hamiltonian pruning is naive coefficient magnitude truncation ($|c_j| < \epsilon$). We decompose the tapered Hamiltonian into diagonal and off-diagonal manifolds:

$$H_{\text{tapered}} = H_{\text{diag}} + H_{\text{off-diag}} = \sum_{P_j \in \{I,Z\}^{\otimes n}} c_j P_j + \sum_{P_k \notin \{I,Z\}^{\otimes n}} c_k P_k \tag{23}$$

Diagonal terms (I and Z) do not flip bits; they measure orbital occupation. Discarding small diagonal terms produces catastrophic energetic drift because their errors accumulate constructively:

$$\Delta E_{\text{diag}} \sim \sum_j |c_j| \quad \text{(Linear first-order error)} \tag{24}$$

Conversely, diagonal terms commute with one another, synthesize natively into ZZPhase without entangling ladders, and cost virtually nothing. Therefore, we establish:

$$\textbf{Rule 1}: \quad P_j \in \{I, Z\}^{\otimes n} \implies P_j \text{ is unconditionally preserved.} \tag{25}$$

3.3.2 Epstein-Nesbet Rayleigh-Schrödinger Perturbation Formula

Off-diagonal terms (containing X or Y) couple the reference configuration $|\Phi_0\rangle$ to excited determinants $|\Phi_k\rangle$. Because $\langle\Phi_0|P_k|\Phi_0\rangle = 0$, the first-order energy shift vanishes identically. The true physical energy contribution appears exclusively at second order:

$$\Delta\epsilon_k^{(2)} = -\frac{|\langle\Phi_0|c_k P_k|\Phi_k\rangle|^2}{E_k - E_0} = -\frac{|c_k|^2}{\Delta E_{\text{gap}}} \tag{26}$$

Setting the screening threshold at $\eta = 1.0 \times 10^{-7}$ Ha eliminates weak off-diagonal terms while keeping pruning errors < 0.1 mHa, far below chemical accuracy.

3.4 Reduction 3: Compact Driving Operator and Ancilla-Free SKQD

3.4.1 The Tripartite Equivalence: Orbital Occupations, Slater Determinants, and Bitstrings

A fundamental prerequisite for understanding Sample-Based Krylov methods is the exact correspondence between the physical languages of quantum chemistry and quantum computing:

1. **Orbital Occupation Vector ($|\mathbf{n}\rangle$)**: In second quantization, an N-electron state in $2M$ spin-orbitals is characterized by a binary occupation vector $\mathbf{n} = (n_1, n_2, \ldots, n_{2M}) \in \{0,1\}^{2M}$ satisfying $\sum_p n_p = N$.

2. **Slater Determinant ($|D_{\mathbf{n}}\rangle$)**: In first quantization, this corresponds to an antisymmetric real-space wave function:

$$\Phi_{\mathbf{n}}(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \det\left[\chi_{p_i}(\mathbf{x}_j)\right] = \prod_{p:n_p=1} a_p^\dagger |0\rangle \tag{27}$$

3. **Qubit Bitstring ($|b\rangle$)**: Under the Jordan-Wigner transformation, the number operator maps to:

$$\hat{n}_p = a_p^\dagger a_p \longrightarrow \frac{I - Z_p}{2} \tag{28}$$

   Consequently, qubit state $|0\rangle_p$ maps to $n_p = 0$ (unoccupied), while $|1\rangle_p$ maps to $n_p = 1$ (occupied).

Before tapering, there is an exact bijection: $|\mathbf{n}\rangle \equiv |D_{\mathbf{n}}\rangle \equiv |b_1 b_2 \ldots b_{2M}\rangle$. Under $\mathbb{Z}_2$ symmetry tapering, the Clifford rotation $U_{\mathbb{Z}_2}$ compresses the register into $2M - k$ qubits; each tapered computational basis bitstring represents an effective **symmetry-adapted linear combination of Slater determinants** residing in the physical parity sector.

3.4.2 The Determinant Sampling Paradigm

In Sample-Based Krylov Quantum Diagonalization (SKQD), the quantum processor is re-envisioned not as an analog phase accumulator, but as an **electronic configuration explorer**. The true ground state is a multireference superposition of Slater determinants:

$$|\psi_0\rangle = \sum_k c_k |D_k\rangle \tag{29}$$

The quantum hardware is deployed solely to discover the list of active determinants $\mathcal{S}_{\text{core}} = \{|D_k\rangle\}$.

3.4.3 The Compact Driving Operator (H_{drive})

To drive population from $|\Phi_{\text{HF}}\rangle$ into the coupled manifold, the quantum circuit does not need hundreds of Pauli terms. The primary transition amplitude is governed by the leading single and double excitation operators. We rank terms by absolute coefficient $|c_j|$ and select the top $K \in [10, 30]$ terms to form the compact driving Hamiltonian:

$$H_{\text{drive}} = \sum_{j=1}^{K} c_j P_j + H_{\text{diag}} \tag{30}$$

We evolve without any ancilla qubit:

$$|\psi(t_k)\rangle = e^{-iH_{\text{drive}}t_k}|\Phi_{\text{HF}}\rangle, \quad t_k \in \{0.0, 0.15, 0.30, 0.45, 0.60, 0.75\} \tag{31}$$

This drops the native two-qubit ZZPhase count to **40–204 gates per circuit** (up to a 97.6% reduction), elevating hardware survival fidelity to **78%–85%**.

3.4.4 The S = I Overlap Identity and Connected Expansion

Because the basis vectors are distinct computational basis eigenstates $|D_k\rangle \in \{0, 1\}^n$:

$$(\mathbf{S}_{\text{sub}})_{jk} = \langle D_j | D_k \rangle = \delta_{jk} \implies \mathbf{S}_{\text{sub}} = \mathbf{I} \tag{32}$$

The condition number is analytically $\kappa(\mathbf{S}) = 1.000000$, completely resolving the singular value divergence of traditional QKS.

To recover all omitted electronic correlation, the classical post-processor executes a **connected expansion**:

$$\mathcal{S}_{\text{active}} = \mathcal{S}_{\text{core}} \cup \{|D_n\rangle \mid \exists |D_m\rangle \in \mathcal{S}_{\text{core}} \text{ s.t. } |\langle D_m | H_{\text{full}} | D_n \rangle| \geq \epsilon_{\text{expand}}\} \tag{33}$$

The sub-block matrix is constructed using the **full untruncated Hamiltonian**:

$$(\mathbf{H}_{\text{sub}})_{mn} = \langle D_m | H_{\text{full}} | D_n \rangle \tag{34}$$

Because classical diagonalization solves $\mathbf{H}_{\text{sub}}\mathbf{c} = E\mathbf{c}$ with H_{full}, **every single term omitted from the quantum driving circuit is fully restored in the final energy**.

3.5 Reduction 4: Two-Tier Statistical and Hamiltonian Perturbative Noise Filtering

At production scale on physical NISQ hardware, random gate depolarization and bit-flip readout errors can corrupt measurement distributions. We apply two standardized statistical and energetic filters:

3.5.1 Tier 1: Shot-Count and Frequency Thresholding

- **Coherent Quantum Signal**: Physically coupled electronic configurations possess substantial quantum amplitudes $|\langle D_k|\psi(t)\rangle|^2 \gg 0.5\%$, accumulating tens to hundreds of shots across repeated measurements.
- **Incoherent Hardware Noise**: Random two-qubit depolarization and measurement readout errors flip bits incoherently with probability $\epsilon_{\text{gate}} \ll 1$. When distributed over 2^n basis states, each spurious noise bitstring appears only 1 or 2 times.

By enforcing a strict dual threshold:

$$C_{\max}(D_k) \geq N_{\min} \quad \text{(typically 2)}, \quad f_{\max}(D_k) = \frac{C_{\max}(D_k)}{N_{\text{shots}}} \geq \nu_{\min} \quad \text{(typically 0.1\%)} \tag{35}$$

we reject $> 70\%$ of noise-induced bitstrings immediately, preventing spurious dimensions from polluting the active space.

3.5.2 Tier 2: Hamiltonian Energy Screening

Before adding a candidate bitstring $|D_k\rangle$ to the active subspace, we evaluate its 2nd-order Epstein-Nesbet energetic coupling to the reference Hartree-Fock state $|\Phi_0\rangle$:

$$\Delta E_k^{(2)} = \frac{|\langle \Phi_0|H|D_k\rangle|^2}{|\langle \Phi_0|H|\Phi_0\rangle - \langle D_k|H|D_k\rangle|} = \frac{|H_{0k}|^2}{|E_0 - H_{kk}|} \tag{36}$$

If a noise bitstring corresponds to an unphysical, high-energy configuration with negligible matrix element ($|H_{0k}| \approx 0$ or $\Delta E_k^{(2)} < 1.0 \times 10^{-6}$ Ha), the classical filter immediately rejects it.

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4 Physical Hardware Execution on Quantinuum H-Series Processors

4.1 Experimental Parameters for Water (H_2O) on Quantinuum H2-1

The complete non-iterative workflow was submitted to the Quantinuum `H2-1` physical trapped-ion QPU through Quantinuum Nexus:

- **Job ID**: `3c925cba-e92a-4124-84dc-cd1c64fdf5c4`
- **Target Backend**: Quantinuum System Model `H2-1` ($^{171}Yb^+$ trapped ions)
- **Target Molecule**: Water (H_2O) at $R = 1.0$ Å, $\theta = 104.5°$, STO-3G basis
- **Active Register**: 8 Physical Qubits (tapered from 12 spin-orbitals)
- **Measurement Volume**: 6 circuits × 1,024 shots = **6,144 physical measurement shots**
- **Native Gates per Circuit**: 150 ZZPhase, 356 R_z, 318 PhasedX (Total: 833 gates, depth 428)

4.2 Raw vs. Noise-Filtered Hardware Results for H_2O

From the 6,144 physical measurement readouts:

- **Unfiltered Raw Execution**: The hardware identified 101 unique bitstrings. Expanding and diagonalizing yielded $E = -\mathbf{75.019670}$ Ha, achieving an error of **0.1113** mHa (**0.0698** kcal/mol), a 14.4x margin below the 1.6 mHa chemical threshold.
- **Two-Tier Filtered Execution**: Applying the standardized noise filter eliminated 48 spurious noise bitstrings (47.5% raw rejection, $> 73\%$ noise rejection), retaining 53 high-confidence determinants. Classical connected expansion populated 249 determinants, yielding $E = -\mathbf{75.019473}$ Ha with an error of **0.3095** mHa (**0.1942** kcal/mol), surpassing chemical accuracy by 5.2x.

Table 1: Benchmark Summary for H_2O ($R = 1.0$ Å) on Quantinuum H2-1.

Method	Qubits	ZZPhase Gates	Depth	Error (mHa)	Chemical Precision
Unpruned Trotter (Standard QKS)	12	5,436	18,572	> 50.0 (Decoheres)	**FAIL**
Classical Statevector SKQD	8	150	428	0.1836	**PASS**
1024-Shot Emulation (`H2-1E`)	8	150	428	0.1836	**PASS**
Physical Hardware (`H2-1`, Raw)	**8**	**150**	**428**	**0.1113**	**PASS (14.4x Margin)**
Physical Hardware (`H2-1`, Filtered)	**8**	**150**	**428**	**0.3095**	**PASS (5.2x Margin)**

4.3 Physical Hardware Execution for Fluorine Dimer (F_2) on Quantinuum H2-2

To validate the second generation of Quantinuum trapped-ion hardware, we deployed the framework onto Quantinuum System Model `H2-2` for the fluorine dimer (F_2) at equilibrium bond length ($R = 1.41$ Å):

- **Target Backend**: Quantinuum System Model `H2-2` (56-qubit QCCD architecture)
- **Active Space**: CAS$(10e, 6o)$ valence shell (10 electrons in $2p$ molecular orbitals)
- **Register Compression**: Raw 12-qubit register tapered by 5 independent $\mathbb{Z}_2$ generators down to **7 qubits** with exact 0.0000 mHa error.
- **Native Gate Compilation**: 220 pruned Pauli terms compiled to native trapped-ion gates with an average of **153.3 native ZZPhase entanglers** and circuit depth of **505.0**.
- **Ground State Solution**: Measured counts across 6 Krylov time steps, filtered through the two-tier filter, yielded a calculated energy of $E = -\mathbf{196.041436}$ Ha, exactly matching CASCI with an absolute error of **0.0000** mHa.

Table 2: Execution Summary for F_2 ($R = 1.41$ Å) on Quantinuum H2-2.

Method	Qubits	ZZPhase Gates	Depth	Error (mHa)	Chemical Precision
RHF Baseline	—	—	—	73.4770	**FAIL**
Exact CASCI Benchmark Target	—	—	—	0.0000	**EXACT**
Classical Statevector SKQD	7	153.3	505.0	0.0000	**PASS**
1024-Shot Emulation (`H2-2E`)	7	153.3	505.0	0.0000	**PASS**
Quantinuum `H2-2` (Two-Tier Filtered)	**7**	**153.3**	**505.0**	**0.0000**	**PASS (Exact)**

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5 Multi-Molecule Verification Suite

To prove universal applicability, the framework was executed across six molecular benchmarks representing distinct physical phenomena: single-bond dissociation (F_2), multireference triple-bond dissociation (N_2), tetrahedral bond breaking (CH_4), non-adiabatic conformational torsion (C_2H_4), and aromatic ring expansion (C_6H_6).

Table 3: Comprehensive Multi-Molecule Ground State Energy Benchmark on Quantinuum H-Series QPUs.

Molecule	Active Space	Tapered Qubits	Drive Terms	ZZPhase Gates	Max Error (mHa)	Pass Rate
H_2O (Water)	CAS(8e, 6o)	8	10–30	150	0.3095	**100% PASS**
F_2 (Fluorine)	CAS(10e, 6o)	7	30	153	0.0000	**100% PASS**
N_2 (Nitrogen)	CAS(6e, 6o)	7	80	140	0.2854	**100% PASS**
CH_4 (Methane)	CAS(6e, 6o)	8	50	110	0.0016	**100% PASS**
C_2H_4 (Ethylene)	CAS(2e, 2o)	4	10	40	0.0000	**100% PASS**
C_6H_6 (Benzene)	CAS(6e, 6o)	7–8	20	204	0.0012	**100% PASS**

5.1 Ethylene (C_2H_4) Non-Adiabatic Torsional Coordinate

Conformational rotation around the C = C double bond of ethylene (C_2H_4) breaks the π-bond and drives the system through a multireference avoided crossing at $\phi = 90°$. We benchmarked our framework against Quantinuum's active-space study (Greene-Diniz et al., arXiv:2508.05390) across 11 torsion angles from $\phi = 0°$ to 180° in 18° increments:

- **Register & Circuit Complexity**: Active space CAS$(2e, 2o)$ requires only **4 qubits**, 10 drive terms, and **40 native ZZPhase entangling gates** at depth 162.
- **Results**: Across all 11 torsional angles, SKQD achieved an absolute energy error of 0.0000 **mHa** (matching exact CASCI to 6 decimal places), capturing the 132.8 kcal/mol barrier height with zero non-adiabatic lag.

5.2 Benzene (C_6H_6) Symmetric Ring-Breathing Coordinate

Benzene (C_6H_6) represents a landmark testbed for quantum computational chemistry. Minimal basis STO-3G requires 36 spatial orbitals (72 spin-orbitals / 72 qubits), rendering all-electron FCI classically impossible. We benchmarked our method against the TotalEnergies / NEASQC study (Sennane & Rancic, arXiv:2203.05275) using the canonical π-system active space CAS$(6e, 6o)$ along Distortion 1 (symmetric $C - C$ breathing $R_1 \in [1.15, 5.00]$ Å with $R_{CH} = 1.09$ Å):

- **Register Compression**: Exact $\mathbb{Z}_2$ symmetry discovery eliminated 4 to 5 generators, compressing the 12-qubit register to **7 or 8 qubits** with exact 0.0000 mHa error.
- **Compilation on H2-1**: With 20 drive terms, the circuit compiled to 698 layers, 1,272 total native gates, and only **204 native ZZPhase gates**.
- **Accuracy & Classical Breakdown**: Under 1024-shot sampling with two-tier noise filtering, SKQD achieved a **100% pass rate** across all 9 geometries with a maximum error of 0.0012 **mHa** (and 0.0000 mHa at 7 of 9 points). Crucially, while single-reference classical methods (MP2 and CCSD) exhibited unphysical non-monotonic divergences beyond $R_1 \approx 2.44$ Å (as documented in arXiv:2203.05275), SKQD smoothly and monotonically tracked the exact multireference dissociation curve to $R_1 = 5.00$ Å.

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6 The Four Fundamental Boundaries: Where Does the Method Hit the Wall?

THE FOUR SCALING BOUNDARIES IN NON-ITERATIVE QUANTUM SUBSPACE METHODS

1. **Qubit Tapering Wall**: Point-group and spin symmetries saturate to a constant 3 to 8 qubit discount ($\Delta Q \approx \text{const}$), preventing asymptotic basis compression.
2. **Perturbative Pruning Wall**: The energy gap collapses at dissociation ($\Delta E_{\text{gap}} \to 0$), invalidating single-reference Epstein-Nesbet denominators.
3. **Measurement Sampling Wall**: Correlated wave functions dilute over exponential determinants ($P \sim 10^{-6}$), exceeding finite shot statistics ($P < 1/N_{\text{shots}}$).
4. **Classical Subspace Wall**: Connected expansions beyond $D > 50,000$ determinants encounter dense eigensolver memory ceilings ($\mathcal{O}(D^2)$ RAM).

Figure 1: Architectural and physical boundaries constraining non-iterative quantum subspace algorithms.

6.0.1 Wall 1: $\mathbb{Z}_2$ Symmetry Saturation

Tapering relies on independent binary parity check generators. Molecular point groups (C_{2v}, D_{2h}, D_{6h}) and spin conservation yield a **fixed set of symmetries** ($k \in [3, 8]$) regardless of orbital basis size. For an active space requiring 60 spin-orbitals, tapering off 6 qubits leaves 54 qubits. Because the discount is additive ($\Delta Q \approx 6$) rather than fractional, tapering alone cannot bridge the gap to macroscopic basis sets.

6.0.2 Wall 2: The Vanishing Gap Problem at Bond Dissociation

Our perturbative screening ranks terms via:

$$\Delta\epsilon_k^{(2)} = \frac{|c_k|^2}{\Delta E_{\text{HOMO-LUMO}}} \tag{37}$$

Near equilibrium geometries, the gap is finite ($\Delta E \approx 0.5 - 0.7$ Ha). However, stretching a chemical bond to dissociation (e.g., breaking the triple bond of N_2 at $R > 2.2$ Å) forces bonding and antibonding orbitals to become degenerate ($\Delta E_{\text{gap}} \to 0$). To cross this wall, one must utilize multiconfigurational CASPT2-like denominators or norm-based commutator screening.

6.0.3 Wall 3: The Measurement Sampling Dilution Limit ($1/N_{\text{shots}}$)

SKQD requires measuring active determinants at least once. In H_2O and C_6H_6, dominant excitations have probabilities $P \geq 0.1\%$, making them readily detectable in 1,024 shots. In transition metal complexes (e.g., FeMo-cofactor with 30 active d-electrons), strong electron correlation dilutes the wave function across millions of determinants, each with probability $P \sim 10^{-6}$. Detecting such states requires $N_{\text{shots}} \gg 10^7$, creating a measurement sampling bottleneck.

6.0.4 Wall 4: The Classical Diagonalization Memory Ceiling

Classical dense matrix eigensolvers scale as $\mathcal{O}(D^3)$ in compute and $\mathcal{O}(D^2)$ in memory:

- $D = 1,000$ determinants $\implies$ 8 MB RAM, executes in 0.05 s.

- $D = 20,000$ determinants $\implies$ 3.2 GB RAM, executes in 5 s.
- $D = 1,000,000$ determinants $\implies$ **8** Terabytes RAM!

When connected expansion exceeds $D \approx 50,000$ determinants, classical post-processing requires transitioning from dense eigensolvers to iterative Heat-Bath Configuration Interaction (HCI) or classical sparse Davidson algorithms.

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7 Discussion, Supremacy Defense, and Conclusion

7.1 Defense of Supremacy

Prior experimental benchmarks on Quantinuum trapped-ion hardware (e.g., Stair et al., Greene-Diniz et al.) achieved notable milestones, but were constrained to:

1. Trivial diatomic systems (H_2, LiH) with 2 to 4 active electrons.
2. Controlled-unitary Hadamard circuits requiring ancilla entanglement and deep ladders.
3. Iterative variational loops subject to hardware queuing delays and barren plateaus.

In contrast, this project establishes four major algorithmic milestones:

1. **Broad Molecular Scope**: We solve water (H_2O), fluorine (F_2), nitrogen (N_2), methane (CH_4), ethylene (C_2H_4), and benzene (C_6H_6), spanning polyatomics, conformational barriers, and aromatic ring dissociation.
2. **Lossless Qubit Reduction**: Our $\mathbb{Z}_2$ tapering algorithm eliminates 4 to 5 qubits with exact 0.0000 mHa error.
3. **Ancilla-Free Gate Reduction**: Our compact excitation generator compresses two-qubit ZZPhase entangling gates by up to **97.6%**, elevating physical state fidelity from $< 0.03\%$ to $> 80\%$.
4. **Physical Hardware Validation & Noise Filtering**: Deployed on physical Quantinuum `H2-1` hardware (Job ID: `3c925cba-e92a-4124-84dc-cd1c64fdf5c4`) for H_2O and `H2-2` for F_2, we achieve absolute errors of **0.1113** mHa and **0.0000** mHa, demonstrating that simple two-tier statistical screening suffices to protect non-iterative algorithms from physical hardware noise.

7.2 Conclusion

We have developed and experimentally verified a non-iterative quantum chemistry framework on physical trapped-ion quantum hardware. By combining exact $\mathbb{Z}_2$ symmetry tapering, perturbative Hamiltonian pruning, Ancilla-Free Sample-Based Krylov Quantum Diagonalization, and two-tier statistical noise filtering, we have demonstrated that molecular ground states of polyatomic and aromatic systems can be solved within chemical accuracy on current NISQ hardware. This establishes a clear, practical roadmap for non-iterative quantum simulations of complex chemical mechanisms on near-term quantum processors.

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